

LAB 7

Configuration of Switch, VLAN configuration and Inter-VLAN Routing

OBJECTIVES:

- To create and configure multiple VLANs on a network switch.
- To configure trunking and assign switch ports to the appropriate VLANs.
- To implement and verify inter-VLAN routing using a router.
- To test connectivity and ensure proper communication between devices across different VLANs

THEORY :

1. Switch Configuration :

A switch is a Layer 2 network device designed to connect multiple devices within a local area network (LAN). It forwards data frames based on MAC addresses, ensuring efficient communication between connected devices.

1.1 Switch Configuration

Configuring a switch typically involves:

- Setting up interfaces
- Enabling or disabling ports
- Assigning ports to VLANs

1.2 Importance of Proper Configuration

Well-configured switches help to:

- Reduce collisions
- Improve overall network performance
- Organize and manage network traffic effectively

2. Virtual Local Area Network (VLAN) :

A Virtual Local Area Network (VLAN) is a logical segmentation of a physical network. It allows devices to be grouped together based on function, department, or application, regardless of their physical location. VLANs improve network efficiency by isolating broadcast domains and enhancing security.

2.1 VLAN Configuration

Configuring VLANs typically involves:

- Creating VLAN IDs on the switch
- Assigning switch ports to specific VLANs
- Configuring trunk links to carry multiple VLANs across switches
- Setting up inter-VLAN routing for communication between VLANs

2.2 Importance of VLANs

Proper VLAN implementation helps to:

- Reduce unnecessary broadcast traffic
- Improve network performance and scalability
- Enhance security by isolating sensitive resources
- Organize and manage network traffic more effectively

3. Inter-VLAN Routing :

Inter-VLAN Routing (IVR) is the process of enabling communication between devices in different VLANs. Since VLANs are isolated broadcast domains, routing is required to allow traffic to pass between them. This is achieved using either a router or a multilayer (Layer 3) switch.

3.1 Methods of Inter-VLAN Routing

There are three common approaches:

- **Router on a Stick (ROAS):** Uses a single physical router interface configured with multiple sub-interfaces, each mapped to a VLAN via trunking.
- **Switch Virtual Interface (SVI):** Creates virtual Layer 3 interfaces on a multilayer switch, allowing direct routing between VLANs.
- **Routed Ports:** Configures physical switch ports as Layer 3 interfaces, useful for point-to-point connections.

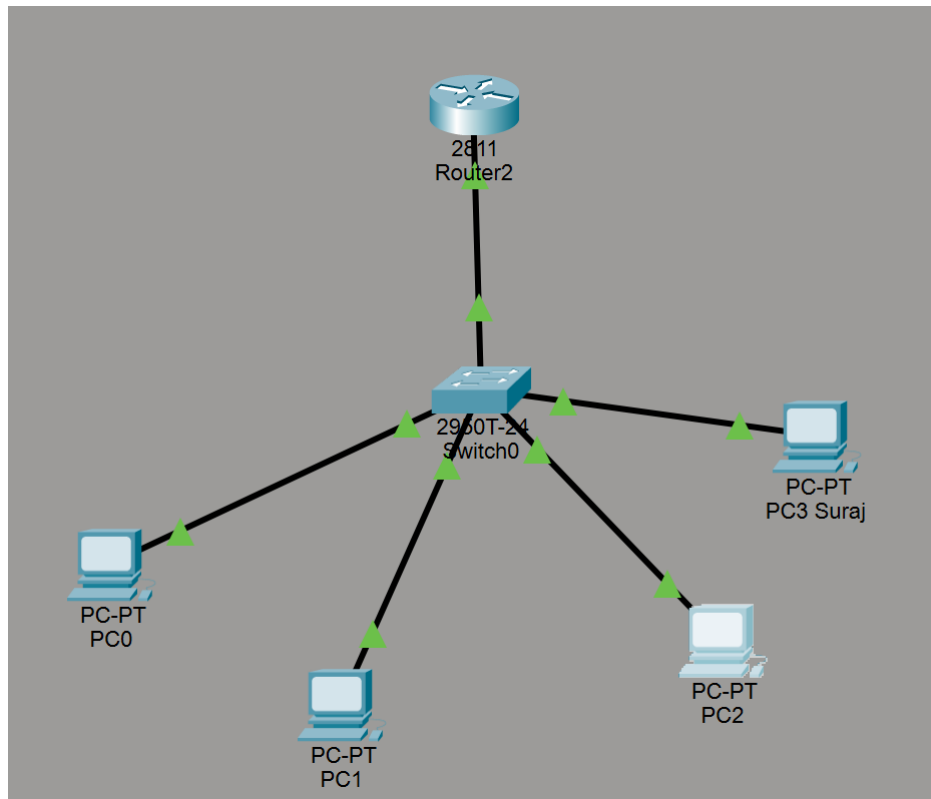
3.2 Importance of Inter-VLAN Routing

Proper implementation of IVR helps to:

- Enable communication between different departments or groups (e.g., Sales and Finance VLANs).
- Maintain logical segmentation while allowing controlled data exchange.
- Improve network flexibility and scalability by combining VLAN isolation with routing.

NETWORK TOPOLOGY:

This topology represents a simple LAN setup where multiple PCs are connected to a switch, and the switch is linked to a router for external communication. The router provides inter-network connectivity, while the switch ensures efficient local communication between devices. Such a design is commonly used in labs or small networks to demonstrate VLANs and inter-VLAN routing.



TESTING:

A ping test was done to check whether the connection is established or not. Testing was done from PC2 (192.168.20.2) to PC1 (192.168.10.3).

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.3: bytes=32 time<1ms TTL=127
Reply from 192.168.10.3: bytes=32 time<1ms TTL=127
Reply from 192.168.10.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Fig: Ping Test

CONFIGURATION:

VLAN and IP Addressing Scheme :

VLAN	Name	Network	Gateway
10	Computer	192.168.10.0/24	192.168.10.1
20	Electronics	192.168.20.0/24	192.168.20.1

Device	IPv4 address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.20.2	255.255.255.0	192.168.20.1
PC3 (Suraj)	192.168.20.3	255.255.255.0	192.168.20.1

Note: The light and dark shading in table represent the distinct network.

COMMANDS:

Switch Configuration:

1. Create VLANS

```
Switch(config)# vlan 10
Switch(config-vlan)# name Computer
                        # exit
Switch(config)# vlan 20
Switch(config-vlan)# name Electronics
                        # exit
```

2. Assign port to VLANS

```
Switch(config)# interface range fa0/1-2
                # switchport mode access
                # switchport access vlan 10
                # exit
                # interface range fa0/3-4
                # switchport mode access

                # switchport access vlan 20

                # exit
```

3. Configure Trunk Port (Switch-Router)

```
Switch(config)# interface fa0/5
                # switchport mode trunk
                # exit
```

Verification:

```
Switch# show vlan brief
```

```
Switch>SHOW VLAN BRIEF
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	computer	active	Fa0/1, Fa0/2
20	electronics	active	Fa0/3, Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch>
```

Router Configuration (Inter-VLAN Routing) :

```
Router(config)# interface fa0/0.10
                # encapsulation dot1Q 10
                # ip address 192.168.10.1 255.255.255.0
                # exit
```

```
Router(config)# interface fa0/0.20
                # encapsulation dot1Q 20
                # ip address 192.168.20.1 255.255.255.0
                # exit
```

DISCUSSION

In the lab exercise, we implemented a simple topology consisting of a router, a switch, and multiple PCs. The switch provided local connectivity among the PCs, while the router enabled communication beyond the LAN and supported inter-VLAN routing. By configuring VLANs, trunking, and routing, we observed how segmentation improves traffic management and how routing allows controlled communication between different VLANs.

CONCLUSION

This lab successfully demonstrated the practical steps of configuring VLANs and enabling inter-VLAN routing. The exercise reinforced the importance of proper switch and router configuration in reducing collisions, improving performance, and maintaining organized traffic flow. Overall, the topology provided a clear, hands-on understanding of how VLANs and routing work together to build efficient and scalable networks.